Monthly Report for ECE FYP/FYT

Project Code:	YH02	Supervisor(s):	Yin Huan
Project Title:	Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)		
Group Member(s):	1) CHAN, Sze Sen 2) XU, Borong		
Reporting Period:	Report #1 ☑ Oct (Fall) Report #2 ☑ Nov (Fall) Report #3 ☐ Feb (Spring) (please attach Reports #1-2 to the Progress Report to be submitted in Jan) (please attach Reports #3 to the Final Report to be submitted in Apr)		
Progress Report: ● List the work completed in this reporting period. ● Identify the major difficulties encountered. ● Comment on the overall progress.	Work completed: ● Calibrated the camera's intrinsic parameters for single-camera localization in the air and under water. ● Calibrated the camera's extrinsic parameters for the new camera array and mount the camera array under the water tank. ● Developed and implemented algorithms to localize ArUco marks. ● Literature review on implementation of RSSI as an assisting localization method. Major difficulties: ● Underwater experiments revealed a gap between in-air and underwater intrinsic calibration. ● A suitable Python environment is yet to be confirmed for the RSSI system, requiring further research. Overall progress: ● Current progress is aligned with our goal, though slightly behind.		
Future Plan: ● Write down the working plan for the next reporting period.	● Investigate the implementation of RSSI as an assisting localization method through literature reviews. ● Use mathematical models to optimize the non-linear point projection from 2D point plane to 3D object point. ● Conduct more underwater experiments to test the algorithm. ● Develop a Python environment showing the relationship between signal strength and position of AUV		
Group Representative's Signature:	Borong Xu		

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